

HENRY HSU

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SUMMARY

Backend and platform engineer building distributed systems and the test infrastructure that keeps them reliable. Built integration test infrastructure and automated failure-triage tooling at Google; optimized 140+ APIs (27.7% faster), scaled async data pipelines from 3M to 20M+ records, and raised test coverage from zero to 67% as a backend engineer and team lead. 14 merged PRs to LlamaIndex (15K+ stars). Pursuing M.C.S. at Rice University.

WORK EXPERIENCE

Software Engineering Intern

Jun. 2026 – Aug. 2026

Google

Taipei, Taiwan

- **Test Infrastructure** – Built 3 integration test infrastructures from a **zero** baseline (**Java, Python**), onboarding **400+** test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Automated Failure Triage** – Built an LLM-backed service that triages scheduled integration test runs and posts root-cause summaries directly onto issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Reproduction Tooling** – Extended the service to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

Back-End Engineer & GenAI Team Lead

Sep. 2024 – Aug. 2025

MaiAgent Co., Ltd. – Award-Winning B2B GenAI Startup

Taipei, Taiwan

- **API Optimization** – Optimized **140+** RESTful APIs through SQL query refactoring, connection pooling (Elasticsearch, Cohere, OpenAI), and Django caching; cut response time of 13 high-traffic APIs by **27.7%** overall and eliminated N+1 queries
- **Data Pipeline & Scaling** – Architected an async processing pipeline with **asyncio, Celery**, and batch/generator patterns for multi-level knowledge base indexing, achieving **5x faster** parsing while scaling from **3M to 20M+** records through a normalized schema and backward-compatible incremental refactors
- **Real-Time Communication** – Implemented a **WebSocket** notification system with **Redis** pub/sub for file parsing status; built event-driven architecture for agent state transitions, pushing render triggers to the frontend for live UI updates
- **CI/CD & Testing** – Built a **GitHub Actions** CI pipeline with pytest unit and E2E coverage, reaching **67%** code coverage from a **zero** baseline; reduced production hotfixes by **90%** initially and **50%** long term; managed database migrations in the CD pipeline
- **Open Source** – Contributed **14** merged pull requests to **LlamaIndex (15K+ stars)**, fixing bugs and adding features across the AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, and MCP Client integrations

EDUCATION

Rice University - Top 20 U.S. University

Aug. 2025 – Dec. 2026 (Expected)

M.C.S., Computer Science (GPA: 4.00/4.00)

Houston, TX

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University

Sep. 2020 – Jun. 2024

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List)

Hsinchu, Taiwan

Minor: Computer Science (Domain GPA: 4.13/4.30)

PROJECTS

Chat Bar - Real-Time Multiplayer Game Server | C/C++ (Socket Programming, SFML), MySQL, SHA-256

Oct. 2023 – Jan. 2024

- **Server-Client Architecture**: Designed and implemented a real-time multiplayer game server using **TCP socket programming** with multi-threaded client handling, supporting concurrent connections for chat and gameplay state synchronization
- **User Management & Client**: Built authentication with SHA-256 password hashing and MySQL persistence plus ranking features backed by optimized queries; developed the SFML client with an event-driven loop for movement, group chat, and live state updates

Elderly Care Map Website | Apache, PHP, MySQL, Google Maps API, Bootstrap5, JavaScript

Sep. 2023 – Jan. 2024

- **Database Design & ETL**: Designed a normalized 8-table schema covering 2,000+ long-term care institutions with joins and constraints for user favorites and geolocation data; integrated government open data (CSV) through an automated ETL pipeline
- **Full-Stack Development**: Built a map-based search platform on the Google Maps API with radius filtering, color-coded bed-capacity indicators, and real-time distance calculation; implemented user authentication and CRUD for saved institutions

AWARDS

Presidential Hackathon Winner (2024, Top 5 Nationally) - Urban noise agent: structured-output LLM pipeline and tool calling

23rd Golden Peak Award (2025) - Outstanding Commercial Product, MaiAgent AI Platform

SKILLS

Languages: Python, Java, C/C++, TypeScript | **Backend**: Django, FastAPI, Express.js | **Frontend**: HTML, CSS, JavaScript, React

APIs: RESTful, WebSocket, Redis Pub/Sub | **Databases**: PostgreSQL, MongoDB, Elasticsearch, Neo4j, Milvus | **Async**: asyncio, Celery

Testing: pytest, JUnit | **DevOps**: GitHub Actions, Docker, git, Nginx, Shell Script | **AWS**: EC2, S3, RDS, ElastiCache | **Methods**: Agile, Scrum